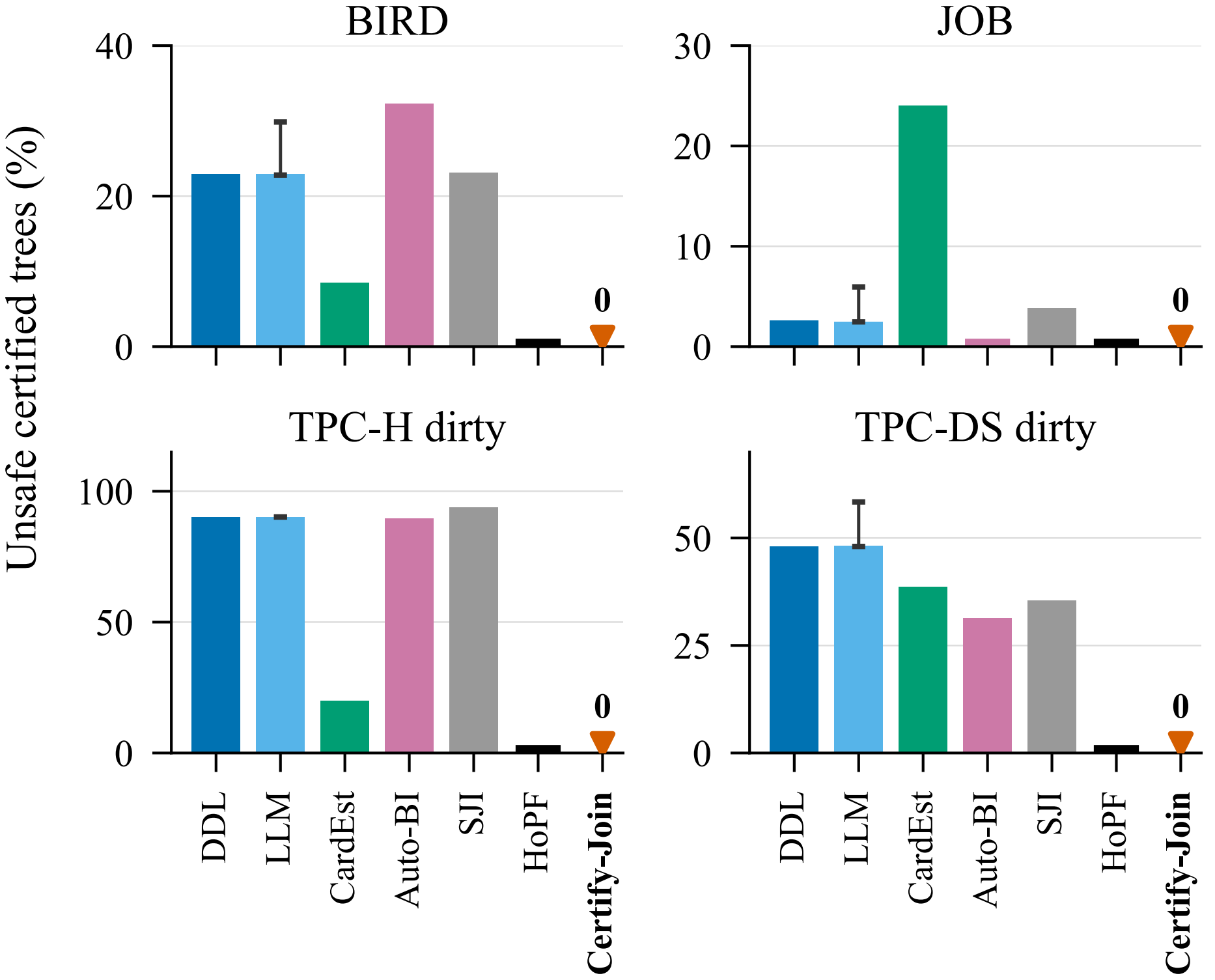
BIRD
JOB
TPC-H dirty
TPC-DS dirty
Unsafe certified trees (%)
40
20
0
30
10
100
50
25
0
DDL
LLM
CardEst
Auto-BI
SJI
HoPF
Certify-Join